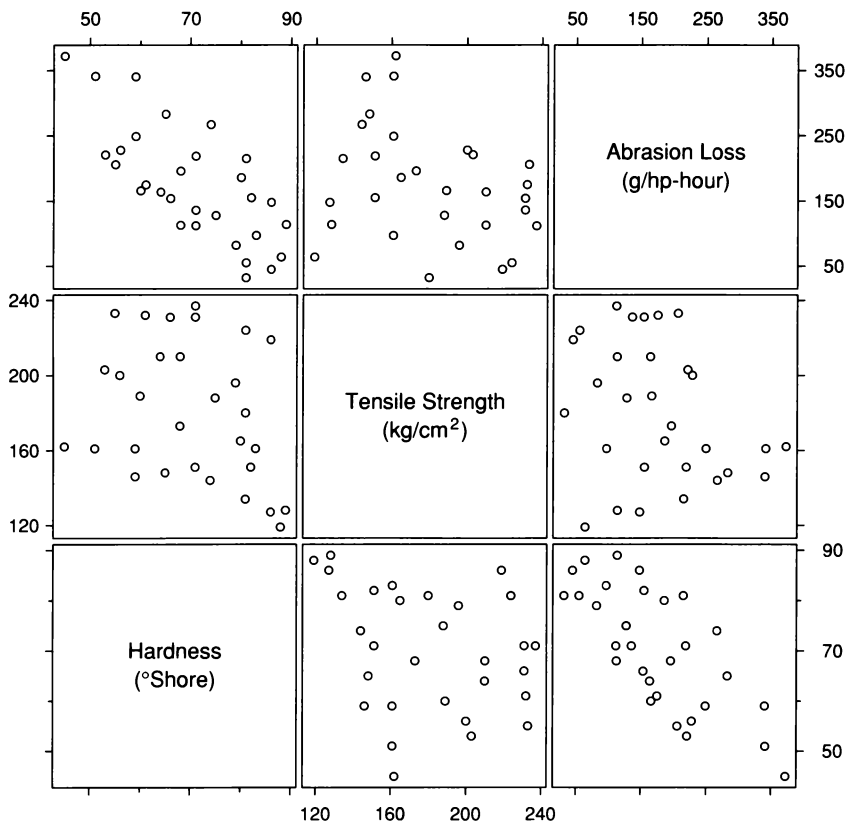




4.1 A scatterplot matrix displays trivariate data: measurements of abrasion loss, hardness, and tensile strength for 30 rubber specimens.

4 *Trivariate Data*

Figure 4.1, a scatterplot matrix, shows data from an industrial experiment in which thirty rubber specimens were rubbed by an abrasive material [31]. Measurements of three variables — abrasion loss, hardness, and tensile strength — were made for each specimen. Abrasion loss is the amount of material abraded from a specimen per unit of energy expended in the rubbing; tensile strength is the force per unit of cross-sectional area required to break a specimen; and hardness is the rebound height of a steel indenter dropped onto a specimen. The goal is to determine the dependence of abrasion loss on tensile strength and hardness; thus abrasion loss is a response, and hardness and tensile strength are factors. Each panel of the scatterplot matrix in Figure 4.1 is a scatterplot of one variable against another. For example, the upper left panel is a scatterplot of abrasion loss on the vertical scale against hardness on the horizontal scale. This scatterplot also appears in the lower right panel, but with the axes reversed.

The rubber data are trivariate data: measurements of three quantitative variables. Geometrically, the 30 values of abrasion loss, hardness, and tensile strength are 30 points in a three-dimensional space. Visualization, whether on a computer screen or on paper, uses a two-dimensional physical medium. The tools in this chapter enable us to visualize the three-dimensional structure within this two-dimensional medium.

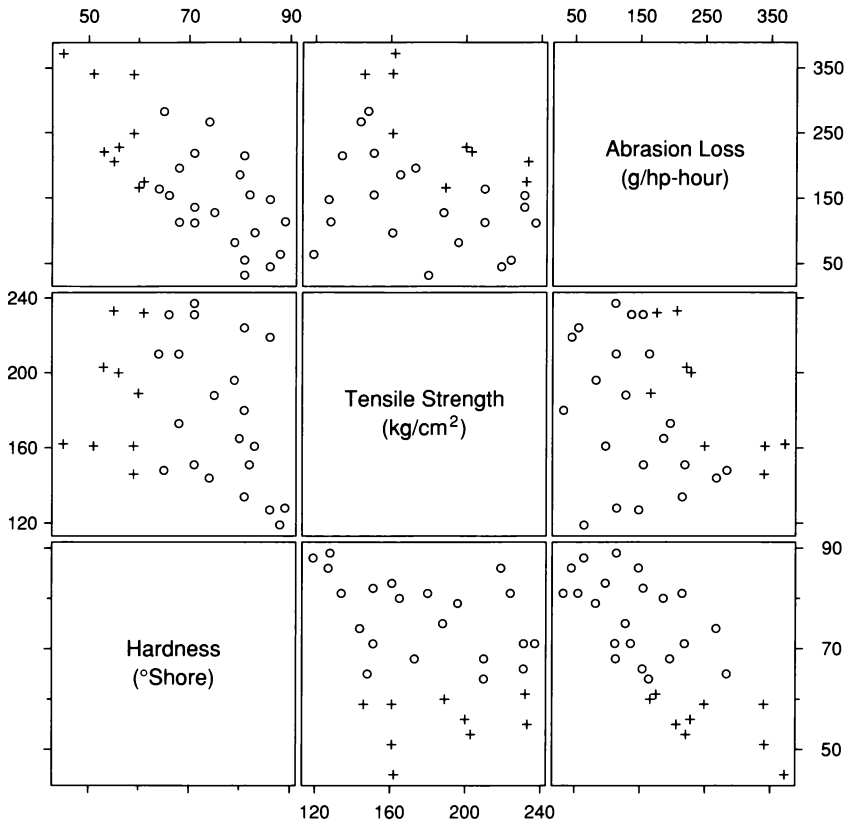
A Coordinate System for Panels

Each panel of a scatterplot matrix, or any other multi-panel display, will be referenced by its column and row numbers; the left column is column one and the bottom row is row one. In Figure 4.1, the upper left panel is (1,3) and the lower right panel is (3,1). This convention amounts to a coordinate system for the panels where the column number is the horizontal coordinate and the row number is the vertical coordinate.

4.1 Coplots of Data

The hardnesses of the rubber specimens range from 45 °Shore to 89 °Shore. Consider those specimens whose hardnesses lie in the range 45 °Shore to 62 °Shore. There are nine such specimens; their data are displayed in Figure 4.2 by the “+” plotting symbols. Geometrically, we have sliced through the three-dimensional space of the data with two planes that intersect the h axis perpendicularly at 45 °Shore and 62 °Shore, and have selected the specimens that lie between the two planes. On the (2,3) panel of Figure 4.2, the “+” symbols show how abrasion loss depends on tensile strength conditional on hardness lying between 45 °Shore to 62 °Shore. There is a nonlinear conditional dependence.

The concept of *conditioning* illustrated in Figure 4.2 is a fundamental one that forms the basis of a number of graphical methods developed in the past [71, 78, 81]. And it forms the basis for the *conditioning plot*, or *coplot*, a particularly powerful visualization tool for studying how a response depends on two or more factors.



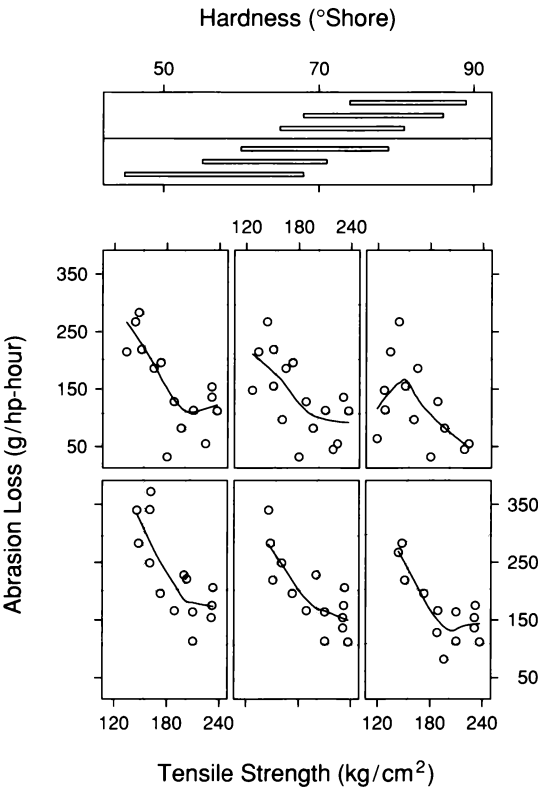
4.2 The “+” plotting symbols encode the data for those specimens with hardness less than 62 °Shore.

The Coplot Display Method

The display method of the coplot presents conditional dependence in a visually efficient way. Figure 4.3 illustrates the method using the rubber data. The panel at the top is the *given panel*; the panels below are the *dependence panels*. Each rectangle on the given panel specifies an interval of values of hardness. On a corresponding dependence panel, abrasion loss is graphed against tensile strength for those observations whose values of hardness lie in the interval. If we start at the (1,1) dependence panel and move in graphical order — that is, from left to right in the bottom row, then from left to right in the next row, and so forth — the corresponding intervals of the given panel proceed from left to right and from bottom to top in the same fashion. The intervals were determined by the equal-count algorithm that was used for slicing in Chapter 3. The use of the algorithm for conditioning will be discussed shortly.

In Figure 4.3, a loess curve has been added to each dependence panel. Bisquare is used in the fitting to protect against outliers that might be present in the data. At this early stage of the analysis, outliers cannot be ruled out, so we take the conservative route of using robust fitting just in case they are there.

The aspect ratio of the dependence panels is critical. In Chapter 3, the aspect ratio was chosen on many displays to bank curves to 45° . In those examples, there was just a single panel. On coplots there are many panels, but they all have the same scales, so we simply apply 45° banking to the entire collection of segments. Suppose all of the symbols and line segments on the dependence panels of Figure 4.3 were graphed on a single panel. The aspect ratio that would result from applying 45° banking to the line segments is the aspect ratio used in Figure 4.3. In other words, the method of 45° banking does not depend on whether we happen to juxtapose subsets of the data on different panels or graph everything on the same panel. And as with single-panel displays, if 45° banking results in an aspect ratio that is too large or too small to maintain the resolution of labels and plotting symbols, we simply use the closest aspect ratio that allows sufficient resolution.

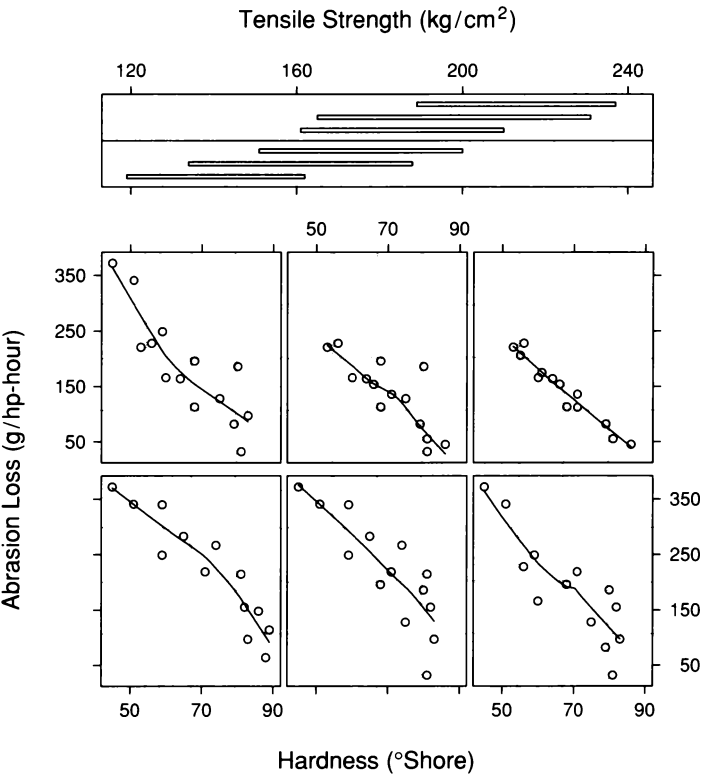


4.3 A coplot graphs abrasion loss against tensile strength given hardness. The parameters of the six robust loess curves are $\alpha = 3/4$ and $\lambda = 1$. The aspect ratio of the data rectangle of the dependence panels has been chosen to produce 45° banking of the collection of line segments that make up the curves.

Figure 4.3 shows a wealth of information about the dependence of abrasion loss on tensile strength. Except for panel (3,2), each conditioning on hardness has a nonlinear pattern: a broken-line, or hockey-stick, function. Below 180 kg/cm², the pattern is linear; above this value, it is also linear, but with a different slope. On the five panels, the slopes below the breakpoint are negative and nearly equal. Above the breakpoint, the five slopes are nearly equal to zero. In other words, the patterns shift up and down, but do not appear to change otherwise by a significant amount. This suggests that for the most part there is no *interaction* between the two factors; the effect of tensile strength on abrasion loss is the same for most values of hardness. But Panel (3,2) shows a major departure from the hockey-stick pattern. As tensile strength decreases, the handle of the stick begins to form but then, suddenly, for the lowest three values of tensile strength, the pattern turns precipitously downward.

Coplots intervals must be chosen to compromise between two competing criteria — number of points and resolution. On the one side, their lengths must be sufficiently great that the dependence panels have enough points for effects to be seen; if there are too few points on a dependence panel, noise in the data typically prevents points from coalescing into a meaningful pattern. On the other side, the lengths must be small enough to maintain reasonable resolution; if a conditioning interval is too big, there is a risk of a distorted view if the nature of the dependence changes dramatically as the value of the conditioning factor changes within the interval. The intervals in Figure 4.3 were selected by the equal-count algorithm. The inputs to the algorithm are the number of intervals and the target fraction of values to be shared by successive intervals. In Figure 4.3, the number of intervals is 6 and the target fraction is $3/4$. These two values were the result of experimentation with a variety of values in an attempt to steer a middle course between resolution and number of points.

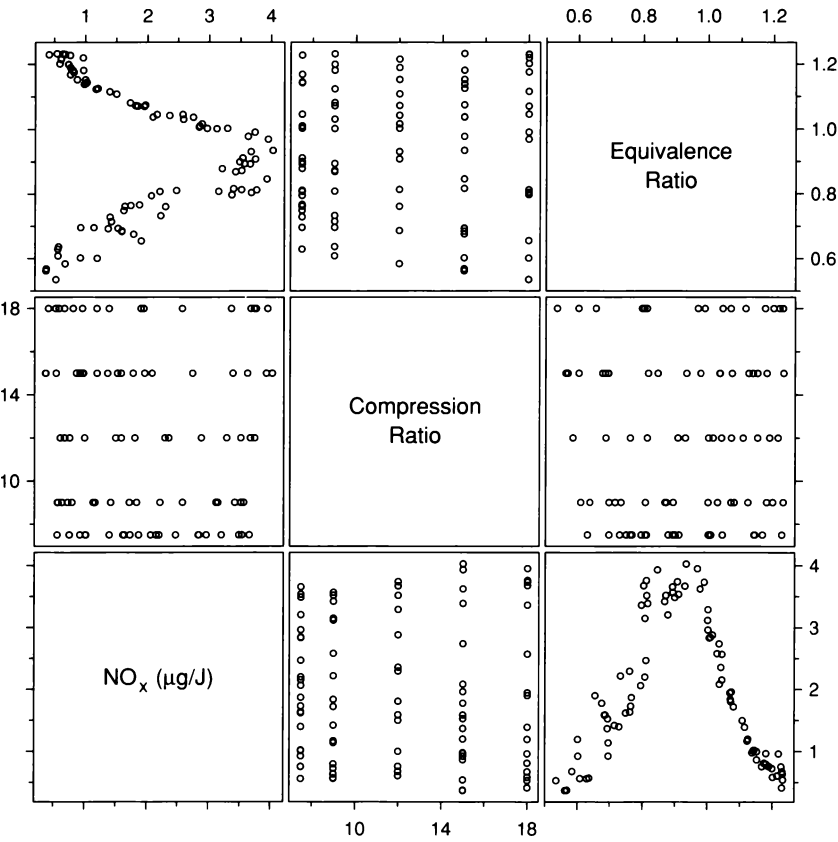
Figure 4.4 conditions on tensile strength; abrasion loss is graphed against hardness for six intervals of tensile strength chosen, again, by the equal-count algorithm with a target fraction of overlap equal to $3/4$. For each conditioning on tensile strength, the dependence of abrasion loss on hardness has, for the most part, a linear pattern. Furthermore, the patterns have roughly the same slope and change only in the intercepts. This supports the observation in Figure 4.3 that, for the most part, there is little or no interaction between hardness and tensile strength. And again, there is a departure from the overall pattern for a small number of points. In panel (1,1), the observations with the three or so largest values of hardness drop well below the linear pattern established by the other observations. The scatterplot matrix in Figure 4.1 shows that the three observations with the smallest values of tensile strength take on the three largest values of hardness. Thus the aberrant behavior on the two coplots is caused by the same three observations, which sit by themselves in a corner of the factor space.



4.4 Abrasion loss is graphed against hardness given tensile strength. The parameters of the robust loess curves are $\alpha = 3/4$ and $\lambda = 1$. The curves are banked to 45° .

Ethanol Data

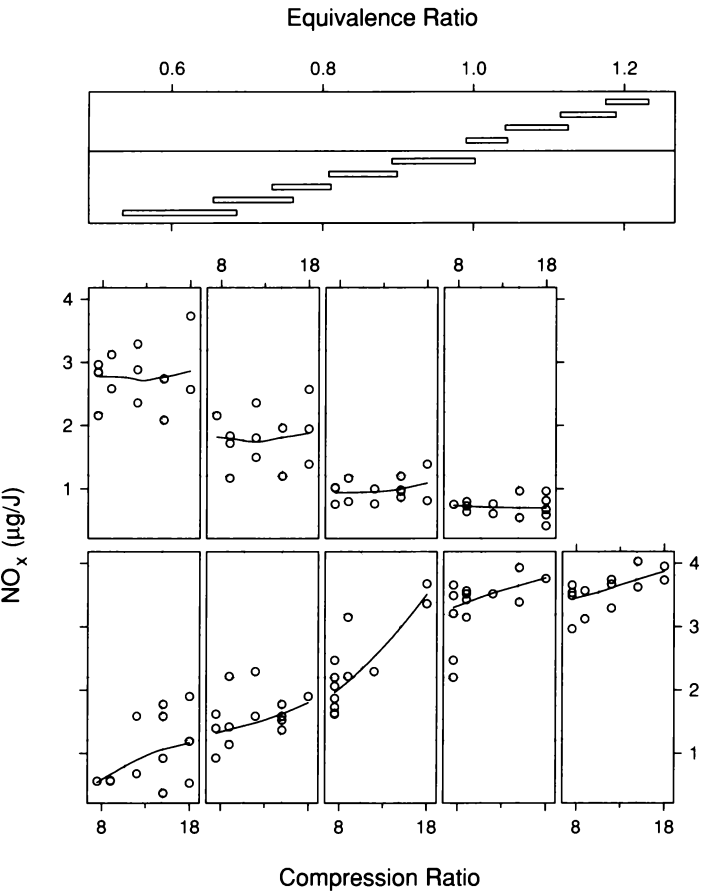
Figure 4.5 is a scatterplot matrix of data from an experiment that studied exhaust from an experimental one-cylinder engine fueled by ethanol [12]. The response, which will be denoted by NO_x , is the concentration of nitric oxide, NO, plus the concentration of nitrogen dioxide, NO_2 , normalized by the amount of work of the engine. The units are μg of NO_x per joule. One factor is the equivalence ratio, E , at which the engine was run. E is a measure of the richness of the air and fuel mixture; as E increases there is more fuel in the mixture. Another factor is C , the compression ratio to which the engine is set. C is the volume inside the cylinder when the piston is retracted, divided by the volume when the piston is at its maximum point of penetration into the cylinder. There were 88 runs of the experiment.



4.5 A scatterplot matrix displays the ethanol data.

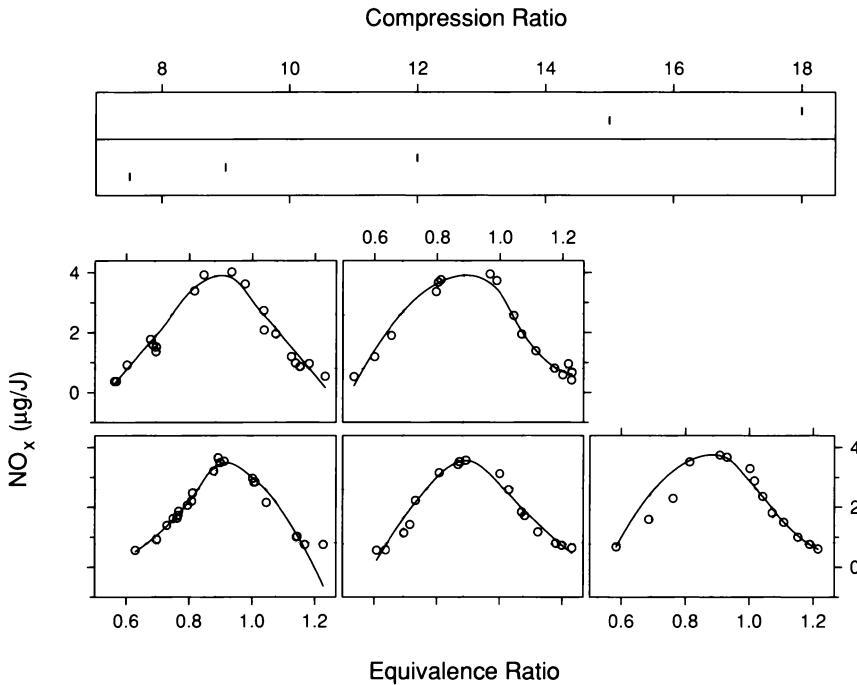
The (3,2) panel in Figure 4.5 shows that C and E are nearly uncorrelated and that C takes on one of five values. The (3,1) panel shows a strong nonlinear dependence of NO_x on E with a peak near $E = 0.9$. The (2,1) panel does not reveal a relationship between NO_x and C , but it is possible that a dependence is being masked by the strong effect of E .

Figure 4.6 is a coplot of NO_x against C given E . The equal-count method has been used to select the intervals; the target fraction of overlap is $1/4$. The coplot shows that NO_x does, in fact, depend on C ; for low values of E , NO_x increases with C , and for medium and high values of E , NO_x is constant as a function of C . In all cases, the underlying pattern appears linear; for low values of E , the slope is positive and for high values, it is zero.



4.6 A coplot graphs NO_x against compression ratio given equivalence ratio. The parameters of the robust loess curves are $\alpha = 1$ and $\lambda = 1$. The curves are banked by an aspect ratio of 2.5; banking to 45° results in an aspect ratio that is too large.

Figure 4.7 is a coplot of NO_x against E given C ; since C takes on only one of five distinct values, each dependence panel corresponds to one of these distinct values. The coplot shows that the peak concentration of NO_x occurs near $E = 0.9$ for all five values of C . But the value of NO_x at the peak increases slightly as C increases. This coplot and the previous one in Figure 4.6 show that the effect of C on NO_x depends on the value of E , and vice versa, so there is an *interaction* between the factors. This contrasts with the rubber data where no major interaction was detected.



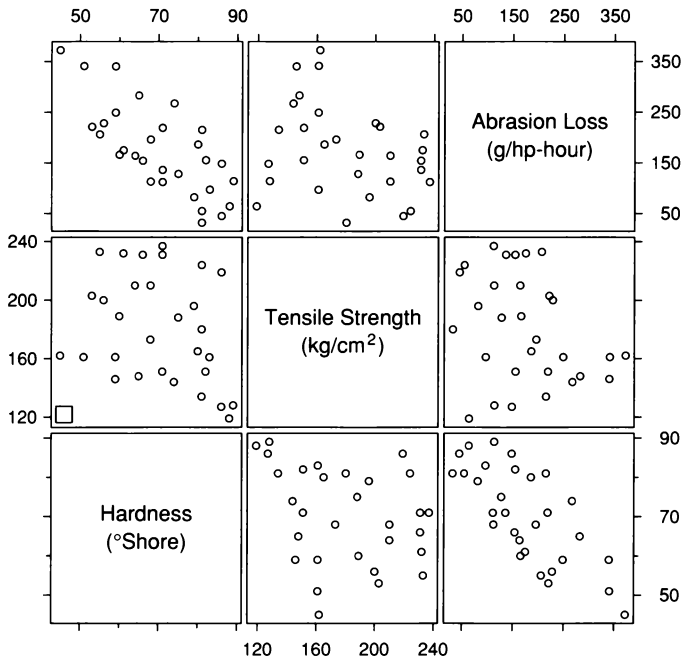
4.7 A coplot graphs NO_x against equivalence ratio given compression ratio. The parameters of the robust loess curves are $\alpha = 3/4$ and $\lambda = 2$. The curves are banked to 45° .

Figure 4.7 shows that the amount of residual variation about the underlying pattern is small. But several observations appear to wander from the underlying pattern by considerably more than the majority of the points. For example, on panel (3,1), two points lie well below the curve. Thus, on the scale of this residual variation, there appear to be outliers, or more pervasive leptokurtosis.

4.2 Direct Manipulation: Conditioning by Brushing

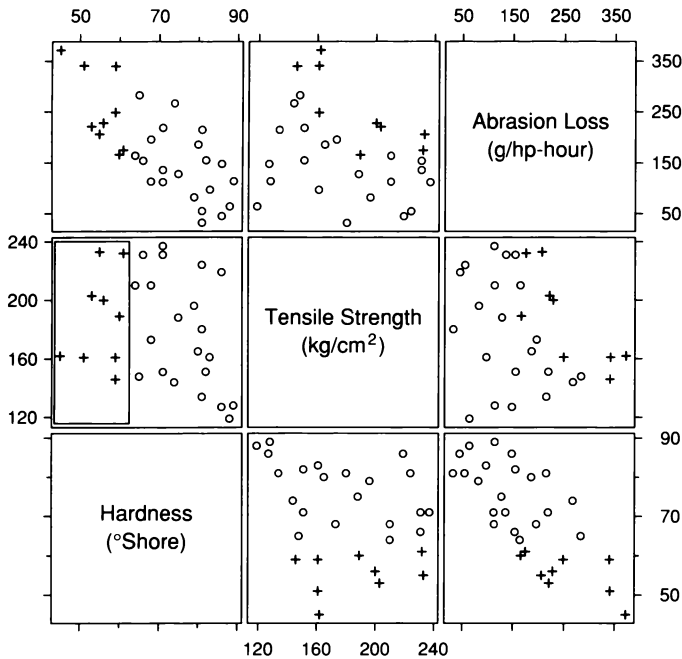
Conditioning analyses can also be carried out by brushing, the direct manipulation tool introduced in Chapter 3. Brushing has a number of *operations* that are applied either in the transient, lasting, or erasing modes. One operation, described in Chapter 3, is labeling. Another, *enhanced linking*, can be used to condition.

Figure 4.8 is a scatterplot matrix of the rubber data. The brush is in the (1,2) panel, and is small and square. First, we turn on the enhanced linking operation of brushing by a menu selection. Next, we change the size and shape of the brush and cover a set of points as shown in



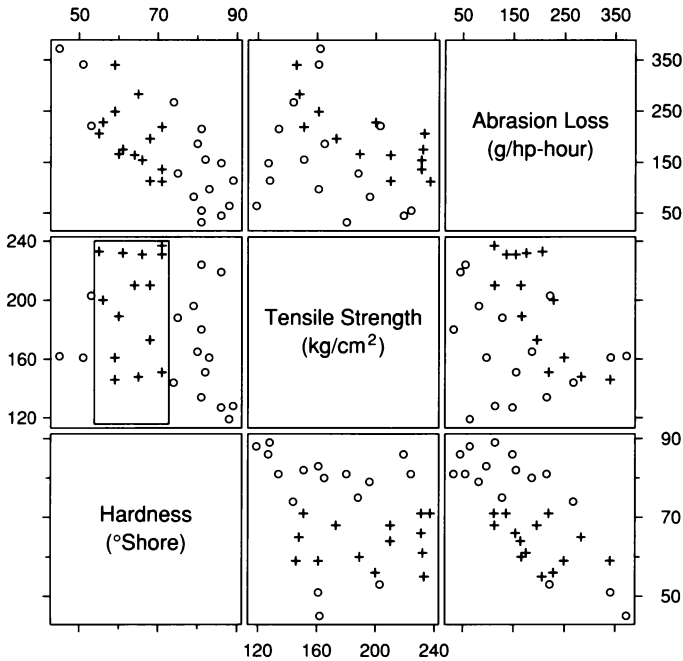
4.8 A scatterplot matrix displays the rubber data. A brush is sitting in the (1,2) panel waiting for instructions to carry out conditioning.

Figure 4.9. The points inside the brush are highlighted as well as all points on the other scatterplots that correspond to these points. The shape and positioning of the brush is vital to producing a conditioning analysis. Its vertical extent spans the range of values of tensile strength, and its horizontal extent spans an interval of low values of hardness. The highlighted points on the (2,3) panel are a scatterplot of abrasion loss against tensile strength conditional on low values of hardness.

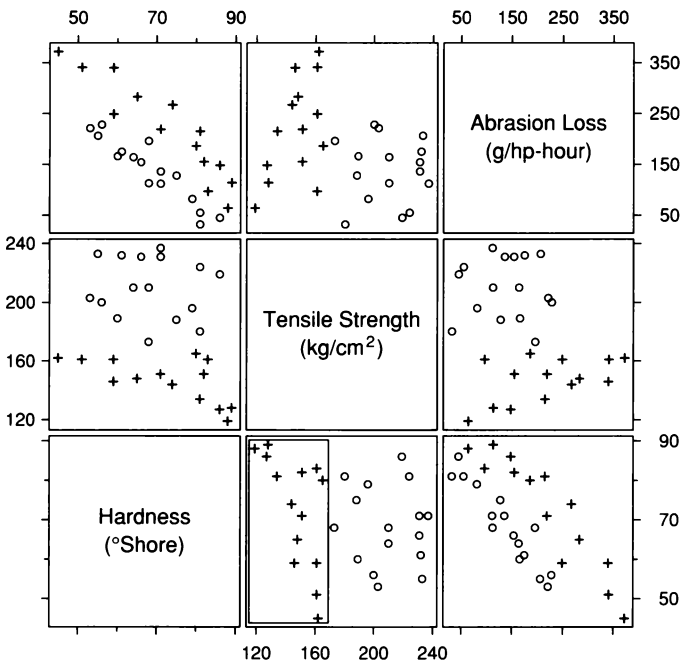


4.9 Brushing with the enhanced linking operation in the transient mode produces a conditioning on low values of hardness.

We can move the brush to the right on the (1,2) panel to condition on other intervals. This is illustrated in Figure 4.10. The mode is transient in this application, so the old highlighting disappears when the brush is moved, and a new set of points is selected for highlighting. Moving the brush left and right on the (1,2) panel provides conditioning on a continuum of hardness intervals. Furthermore, we can quickly change the width of the brush, making it smaller to increase the resolution of the conditioning, or making it larger to decrease the resolution, and then resume the brushing left and right with the new width. Also, moving the brush to the (2,1) panel allows conditioning on tensile strength. This is illustrated in Figure 4.11; the highlighted points on the (1,3) panel show the dependence of abrasion loss on hardness for tensile strength in an interval of low values.



4.10 The brush has been moved to the right to condition on a different interval of hardness.



4.11 The brush has been moved to the (2,1) panel to condition on tensile strength.

The term “enhanced linking” is used for the brushing operation just shown because the scatterplot matrix by itself provides some visual linking of effects on different panels just through visual scanning along rows and along columns. This issue of visual perception is taken up in more detail in Chapter 5.

4.3 Coplots of Fitted Surfaces

In Chapter 3, fitting was introduced to aid the visualization of the dependence of a response, y , on a single factor, x . The two methods were least-squares fitting of parametric families and loess curve fitting. And bisquare iterations were added to these two methods when outliers or more pervasive leptokurtosis was present.

The same approach can be used to study the dependence of y on two factors, u and v . Suppose the data are u_i, v_i , and y_i for $i = 1$ to n . The fitted function is now a surface, $\hat{g}(u, v)$, a function of two variables. To visualize the data, we graph both the fitted function and the residuals,

$$\hat{\varepsilon}_i = y_i - \hat{g}(u_i, v_i) .$$

If the underlying pattern in the data is simple, parametric fitting might be satisfactory. If the underlying pattern is too complex for a parametric fit, then loess can be used.

Parametric Fitting

Least-squares fitting for trivariate data is a simple extension of the method for bivariate data. Suppose the underlying pattern of the dependence of y on u and v is linear. To fit a linear parametric family by least-squares, we find values of a, b , and c that minimize

$$\sum_{i=1}^n (y_i - a - bu_i - cv_i)^2 .$$

This minimizes the sum of squares of the residuals.

Bisquare fitting is also a simple extension. The first step is to fit by least-squares. Then, residuals and bisquare robustness weights, r_i , are

computed, and the function is fitted again using weighted least-squares with weights r_i . For a linear parametric family, the weighted least-squares fit minimizes

$$\sum_{i=1}^n r_i (y_i - a - bu_i - cv_i)^2 .$$

This weighted fitting is then iterated until the fit converges.

Loess

For two factors, loess fitting is similar to the one-factor case; details are given at the end of this section. The loess parameter α controls the amount of smoothness of the fitted surface; as α increases, the smoothness of the surface increases. The loess parameter λ controls the degree of the local fitting. If $\lambda = 1$, the fitting is locally linear, so linear polynomials,

$$a + bu + cv ,$$

are fitted. If $\lambda = 2$, the fitting is locally quadratic, so quadratic polynomials,

$$a + bu + cv + duv + eu^2 + fv^2 ,$$

are fitted. And bisquare iterations can be added when robustness is needed.

Distance, however, must be given additional consideration for two factors. For the case of one factor, x , the neighborhood weight, $w_i(x)$ given to (x_i, y_i) for the fit at x is based on the distance of x_i from x . Since x and x_i are points lying on a line, it is natural to use $|x - x_i|$ as the distance between them. For two factors, the value at which the fit is computed is now a point (u, v) in the plane, and the measurements of the factors are other points (u_i, v_i) in this plane. We could simply take distance from (u, v) to (u_i, v_i) to be Euclidean,

$$\sqrt{(u - u_i)^2 + (v - v_i)^2} .$$

This might be sensible in cases where u and v describe geographical location and thus have the same units. But in most applications of loess, the units of the factors are different. For example, for the rubber data,

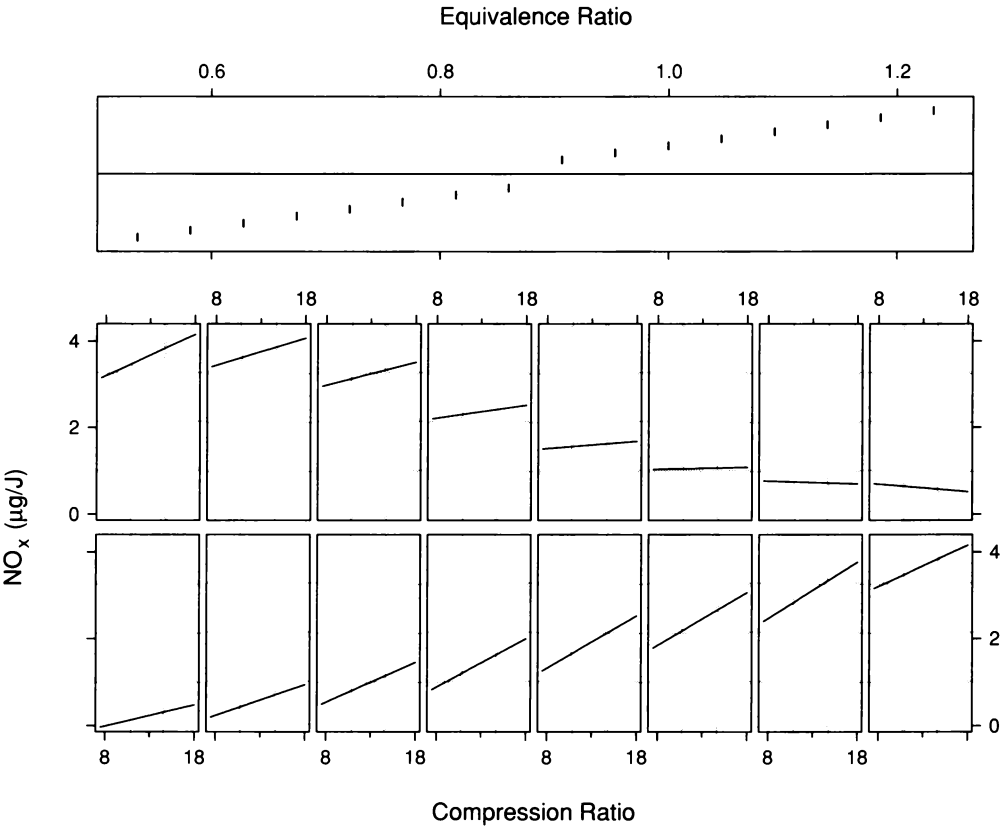
the units of the factors are kg/cm^2 for tensile strength and $^\circ\text{Shore}$ for hardness. Euclidean distance depends on the choice of these units. If we changed from kg/cm^2 to g/cm^2 , all values of tensile strength would be multiplied by 10^3 , and relative values of the Euclidean distances would change drastically. We can remove the dependence on units by dividing each factor by a measure of spread, then using Euclidean distance for the standardized variables. One way to standardize would be to divide each factor by its sample standard deviation. But as we saw in Chapter 2, the sample standard deviation is not robust; it can be badly distorted by just a few outliers that are much smaller or much larger than the majority of the data. To protect against this, we will use a *trimmed sample standard deviation*. The values of the factor are sorted from smallest to largest, the smallest 10% and the largest 10% are dropped, and the sample standard deviation is computed from the remaining 80% of the values.

Coplots of a Fit to the Ethanol Data

For the ethanol data, the coplots in Figures 4.6 and 4.7 showed that the dependence of NO_x on C and E is too complex to be adequately fitted by a simple parametric family. Thus we will use a loess fit, $\hat{g}(C, E)$, to describe how NO_x depends on C and E . Figure 4.7 showed that bisquare is needed in the fitting because some observations deviate substantially from the underlying pattern compared with the deviations of the majority of the observations. Figure 4.7 also showed that conditional on C , NO_x as a function of E has a large amount of curvature with a peak near $E = 0.9$. Because of the curvature, we must take $\lambda = 2$. But Figure 4.6 showed that given E , the underlying pattern in the data is linear in C . By using a feature of loess called *conditionally linear fitting*, we can produce a fitted surface that is linear in C given E . As E changes, the slope and intercept of the fit change, but it is always linear in C given E . The details of conditionally linear fitting are given at the end of this section. Finally, a value of α equal to $1/3$ resulted from trying different values to find one that produced no lack or surplus of fit; this will be discussed further in Section 4.4.

Coplots can also be used to visualize fitted surfaces. Figure 4.12 is a coplot of the robust loess fit, $\hat{g}(C, E)$, to the ethanol data. As with coplots of data, the given panel is at the top and the dependence panels are below. Consider the (1,1) dependence panel. E has been set to a specific conditioning value, 0.535. Then $\hat{g}(C, 0.535)$ has been evaluated for

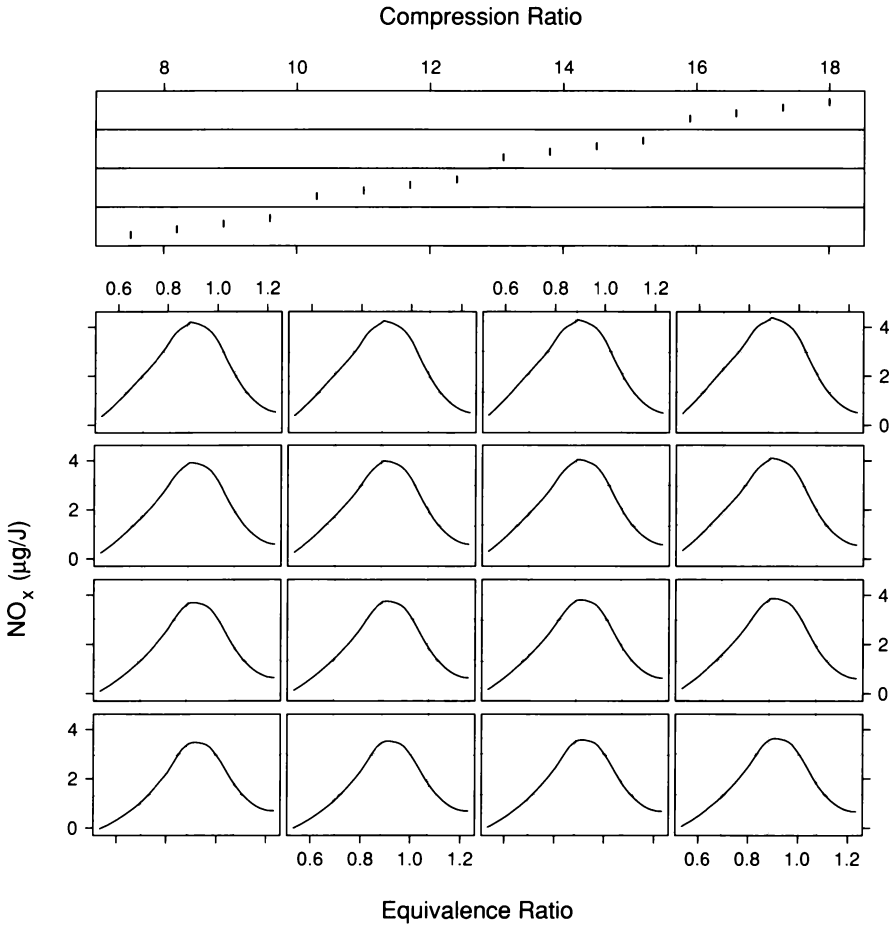
equally spaced values of C ranging from the minimum value of C in the data to the maximum. On the panel, the values of $\hat{g}$ are graphed against the equally spaced values of C . The same method is used on the other dependence panels, but for different conditioning values of E . There are 16 equally spaced conditioning values ranging from the minimum value of E in the data to the maximum; the given panel shows the 16 values.



4.12 A coplot graphs the robust loess fit against compression ratio given equivalence ratio. The curves are banked by an aspect ratio of 2. Banking to 45° results in an aspect ratio that is too large.

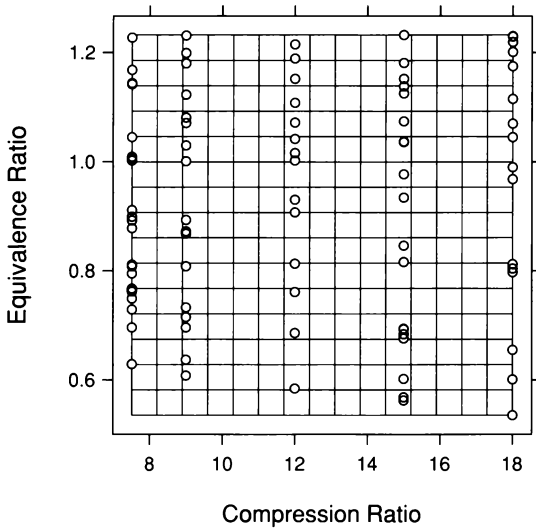
Figure 4.12 shows that given E , $\hat{g}$ is linear in C , the result of specifying the loess fit to be conditionally linear in C . As the conditioning values of E increase from the lowest values, the slopes of the lines first increase until E is about 0.8, and then decrease to zero. Thus, as we saw in the coplots of the data, there is a strong interaction between C and E .

Figure 4.13 is a coplot of the robust loess fit against E for 16 conditioning values of C . Figure 4.13 shows that $\hat{g}$ varies in a highly nonlinear way as a function of E given C . The maximum of $\hat{g}$ increases by about $1 \mu\text{g}/\text{J}$ as the conditioning values of C go from the minimum to the maximum; the value of E at which this maximum occurs is always close to 0.9.



4.13 A coplot graphs the robust loess fit against equivalence ratio given compression ratio. The curves are banded to 45° .

Figure 4.14 illustrates further how the coplots of the fit, $\hat{g}(C, E)$, have been constructed. The circles are the observed values of C and E . The fit is a surface lying above this plane of (C, E) values. Figure 4.14 shows 16 horizontal line segments drawn at the 16 conditioning values of E ; each of these segments ranges in the horizontal direction from the minimum value of C in the data to the maximum. For the coplot of Figure 4.12, the surface is graphed along the 16 horizontal segments. Figure 4.14 also has 16 vertical segments, drawn in the same manner, but showing the conditioning on C . For the coplot of Figure 4.13, the surface is graphed along the 16 vertical line segments.



4.14 The coplots of the robust loess surface display the surface along the 16 horizontal and 16 vertical line segments shown here.

The coplots have provided an effective visualization of the loess surface fitted to the ethanol data. But we must reserve judgment about whether this has shed light on the structure of the data. We must validate the fit by visualizations of the residuals, making sure that the fit has been true to the data. In other words, we must convince ourselves that the coplots have not provided a thorough understanding of nonsense. Validation of the fit will be discussed in Section 4.4.